

Sahil Kumar

Dual Degree (B.Tech ECE – NIT Uttarakhand — BS Data Science – IIT Madras)

✉ kumarsahil9767@gmail.com ☎ +91-7668470448 🌐 Sahil-kumar

Education

Indian Institute of Technology Madras Diploma level - Bachelors in Data Science and Applications CGPA: 7.84 / 10	Chennai, Tamil Nadu May 2023 – Dec-2027
National Institute of Technology Uttarakhand B.Tech - Bachelors in Electronics and Communication Engineering CGPA: 8.58 / 10	Srinagar, Uttarakhand Aug 2023 – May-2027
Shri Balwant Singh Sr. Sec. School Science Stream (CBSE) — Class 12: 96% , Class 10: 90%	Hathras, Uttar Pradesh Mar 2020 – Jun 2022

PROJECTS

Placement Portal System 🔗 Flask • SQLAlchemy • Vue.js • REST API • Redis • Celery - Engineered a scalable full-stack placement management system supporting multiple user roles (Admin, Company, Student) with secure access control. - Architected a modular backend system (services, models, routes) enabling maintainability and ease of feature expansion. - Integrated asynchronous task processing using Celery and Redis to handle email notifications, reminders, and report generation without blocking main application flow. - Built a real-time application tracking workflow with multi-stage status transitions (applied → shortlisted → interview → selected), improving user visibility and engagement. - Developed an ATS-based resume evaluation module, enabling automated comparison of resumes with job descriptions using keyword matching techniques. - Implemented secure authentication mechanisms with password hashing and role-based authorization to enforce data protection and access control. - Designed the system with cloud deployment readiness, leveraging environment-based configuration and scalable architecture principles.	Dec 2025
Virtual Teaching Assistant – RAG-Based LLM Backend 🔗 FastAPI • LangChain • ChromaDB • Docker • REST API - Engineered a Retrieval-Augmented Generation (RAG) backend enabling semantic query resolution over structured course content. - Designed scalable FastAPI-based microservice architecture for handling structured JSON request-response cycles. - Integrated vector database (ChromaDB) for efficient embedding indexing and similarity-based retrieval. - Implemented query preprocessing and context injection pipeline for improved LLM response relevance. - Containerized and deployed the system using Docker for reproducible environment setup and production readiness. - Ensured modular backend structure allowing easy integration with frontend or external systems.	Sep 2025

SKILLS

Languages: Python, C, basic C++, SQL
Backend: Flask, FastAPI, REST APIs, SQLAlchemy
Databases: MySQL, PostgreSQL
AI and framework: Machine learning, Langchain, RAG system
Core CS: Data Structures, Algorithms, OOP, OS, DBMS,
Tools: Docker, Git, GitHub

ACHIEVEMENTS

- Solved 150+ problems on LeetCode covering DSA topics (Arrays, Graphs, DP, Trees).
- Student Placement Representative – NIT Uttarakhand.